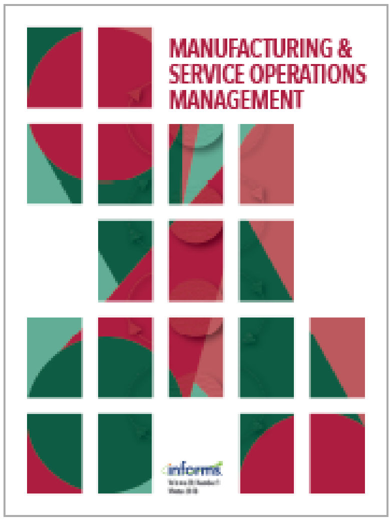


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# Operational Disruptions, Firm Risk, and Control Systems

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**Abstract.** ***Problem definition***: Operational disruptions can impact a firm's risk, which manifests in a host of operational issues, including a higher holding cost for inventory, a higher financing cost for capacity expansion, and a higher perception of the firm's risk among its supply chain partners. ***Academic/practical relevance***: Although disruptions have been studied extensively in the operations management literature, the emphasis has been on mitigating the deleterious impact on the firm's production potential rather than its risk. We examine whether implementing and credibly attesting to having effective internal control systems will meaningfully influence the impact of operational disruptions on the firm's risk and market valuation. ***Methodology***: We exploit a 2004 regulatory change that required certain firms to attest to having control systems in place. Our triple-difference regression specification takes advantage of a regulatory anomaly that excluded a well-defined set of firms from complying with the control system requirement. ***Results***: We find that firms that were obligated to comply with this regulatory requirement experienced a materially smaller increase in their risk and a smaller decrease in their market value in the aftermath of an operational disruption. Firms that were not obligated to comply with this requirement did not experience such benefits in the aftermath of a disruption and instead, experienced a larger increase in their risk. ***Managerial implications***: Fostering a better understanding of whether credible control systems can reduce the impact of disruptions on the firm's risk and value is important as it identifies a broader set of mitigation strategies available to operations managers. This can help managers achieve a better fit between their risk mitigation initiatives and their objectives and budget constraints.

**Supplemental Material:** The online appendix is available at https://doi.org/10.1287/msom.2020.0943.

**Keywords:** operational risk • operational disruptions • information asymmetry • control systems



## 1. Introduction

Operational disruptions remain a major exposure for many companies. In January 2018, the Allianz Risk Barometer survey identified business interruption and supply chain disruption as the number one peril among risk managers and corporate insurance experts for the sixth year in a row (Dobie et al. 2018). Recent academic and industry research confirms that operational disruptions negatively affect firm performance (Sheffi 2005, Simchi-Levi et al. 2015), market value (Hendricks and Singhal 2003, World Economic Forum and Accenture 2013), and risk (Hendricks and Singhal 2005). The market value destruction that is brought on by a disruption can be attributable to both the direct economic consequences of the disruption and investor perceptions that the firm's operations are riskier than previously believed. For instance, on June 13, 2006 the European Aeronautic Defence and Space Company (EADS) announced that an "industrial issue" associated with the "manufacturing and installation of electrical systems and resulting harnesses" would interrupt the production schedule of the Airbus A380 commercial aircraft (Airbus 2006). The investing community noted that this disruption both had a direct financial impact and increased the risk that investors associated with the firm's operations. In a research note, Deutsche Bank stated:

> This news comes [as] a disappointment not only on the A380 directly but also raising concerns and perceived risks on Airbus' ability to manage further large R&D programmes, such as the A350. Factoring in the negative cash flow effect of the A380 slip implies a Euro 3.0-3.5 direct valuation impact to our prior Euro 33 target price. In addition to this, reflecting the likely <u>higher risk premium the market will now apply to EADS following this news</u> [emphasis added], we have increased the WACC [weighted average cost of capital] used in our 10 year DCF [discounted cash flow] model by around 10% (to 9.4%). Overall, this causes our target price to drop by 20% to Euro 26. (Fidler and Memmi 2006, pp. 1)

In this case, the 7€ decrease in the target share price was attributed to a 3–3.5€ direct valuation impact and therefore, a 3.5–4€ increase in the risk premium in the discounted cash flow model. EADS stock lost nearly 30% of its value on the day after the disruption announcement.

In light of the negative consequences that disruptions bring to firms, a growing stream of research examines how firm actions or characteristics may mitigate the economic consequences of disruptions. Much of this work emphasizes changes that operations managers can make to buffers, such as dual sourcing (Tomlin 2006, Wang et al. 2010, Tomlin and Wang 2011), increased safety stock (Tomlin 2006, Tomlin and Wang 2011), inventory levels (Dong and Tomlin 2012), overordering (Dada et al. 2007), and strategic inventory placement (Tang 2006). Other scholarly research shows the conditions under which different disruption mitigation strategies are most effective, including process improvements (Wang et al. 2010, Tomlin and Wang 2011), dynamic pricing (Swinney and Netessine 2009), and demand management (Tomlin and Wang 2011). Empirical studies have found that the occurrence of a disruption is influenced by the frequency of planned maintenance (Guajardo et al. 2012) and that the impact of a disruption on the firm's market value is correlated with the amount of slack in the firm's operations (Hendricks et al. 2009).

It is possible, however, that other actions under an operator's control would allow her to mitigate the deleterious effect of disruptions on the firm. Identifying such actions would be useful as it would provide the operator a larger choice set over which to compose optimal mitigation decisions. As described by Hopp and Spearman (2001, p. 197), "An operations manager has three basic assets to manage to generate return on investment: information, control, and buffers." This observation motivates us to examine a less-studied asset that may benefit operators—internal control systems. In particular, we examine whether firms can mitigate the impact of an operational disruption on the firm's risk and value by implementing and credibly communicating the presence of internal control systems. An internal control system can include systems or processes that allow firms to monitor and manage the deployment of resources (inventory, people, capacity, capital, intellectual property, etc.) to credibly communicate the company's performance. Examples include enterprise risk management systems, business continuity plans, inventory control systems, and enterprise resource planning systems. Although such processes and systems can benefit a company's steady-state performance if they have been effectively implemented, it is not obvious whether a firm can benefit from credibly communicating to external constituents that the firm has effective control systems in place prior to the occurrence of a disruption.

Several difficulties arise in executing a robust empirical examination of the effects of internal control systems on firm outcomes. In an ideal setting, a researcher would randomly assign firms to credibly and effectively implement control systems that are suited to each firms' business needs and compare those outcomes with outcomes from firms that did not receive such random assignment. In practice, however, firms are able to choose whether to implement control systems, when to implement those systems, how effectively they implement the systems, and whether to communicate these choices to outside constituents like investors, suppliers, and customers. This makes empirical analysis susceptible to endogeneity problems as each firm will likely make choices that produce the biggest benefit for their particular circumstances.

We tackle this issue by taking advantage of the enforcement of Section 404 of the Sarbanes-Oxley Act (SOX), an exogenous regulatory shock that occurred during our study period. Section 404 requires that management include an internal control report with the firm's annual report that affirms "the responsibility of management for establishing and maintaining an adequate internal control structure and procedures" and provides an assessment "of the effectiveness of the internal control structure and procedures of the issuer for financial reporting" (107th Congress 2002, pp. 40). As pointed out by Howell et al. (2005), maintaining the control processes required by Section 404 falls to both operations and finance personnel within the firm. For a useful overview of operational and supply chain controls that facilitate compliance with Section 404 of SOX, refer to Protiviti and APICS (2003).

A particularly useful feature of this regulatory mandate is that firms are not required to implement a particular control system, only to attest that whatever system that is put in place ensures "adequate internal structure and procedures." Such control systems will necessarily differ for firms based on their business needs and operating environment. One can imagine that a service provider would have different needs than a manufacturer or a commodity extractor, a make-to-order manufacturer would have different needs from a make-to-stock manufacturer, a firm with a global supply chain would have different needs from a firm with a domestic-only supply chain, and so on. By leveraging this policy shock, we align our analysis with the practical reality that firms should not forcibly implement a generic control system but rather, a control system that is appropriate and effective for their unique needs.

Most firms were required to adhere to the new regulations beginning with the firm's fiscal year ending on or after July 15, 2004 (Securities and Exchange Commission 2003). We take advantage of an idiosyncrasy in the new regulations that excludes firms of a certain size from compliance. This allows us to analyze the market response to disruption announcements made by firms that complied with these control requirements compared with the response to firms that did not comply. For ease of exposition, we sometimes refer to firms that implement and credibly convey the efficacy of their internal control systems as firms "with credible controls" and firms that do not meet this standard as firms "without credible controls." Our setting affords us the unique opportunity to control for many of the endogenous choices that would otherwise plague an analysis of control systems. Those firms that are obligated to adhere to the internal controls requirement must do so by an established time, at an established level of compliance, and communicate their status clearly to investors (and any other interested constituent that wants to read the firm's Securities and Exchange Commission (SEC) filings). This natural experiment brings us close to an ideal experimental setting.

This regulatory change provides a quasiexperimental setting that is well suited to a matched difference-in-difference-in-differences (DDD) analysis. Our DDD model examines the risk and valuation impact of firms depending on whether they are targeted by the control system requirement (first difference), before and after the enforcement date of this requirement (second difference), and whether they experience a disruption or belong to a matched set of firms with a similar propensity to be disrupted (third difference). A model based on the first two differences alone would allow us to compare firms that implement and credibly communicate the presence of control systems to firms that are excluded from complying with this new requirement, both before and after the regulatory enforcement date. This setup explicitly accounts both for common trends in the dependent variables and for observable and unobservable fixed effects, including those that differentiate the firms that comply versus do not comply (such as size). However, such a difference-in-differences (DD) model does not address the possibility that disruptions may occur disproportionately for certain types of firms in times when the stock market is more volatile or trending. If this is true, the occurrence of a disruption is endogenous to the firm's risk and value prior to the disruption. This may create a bias for or against a result in our analysis. We mitigate this concern by creating a matched set of nondisrupted firms that are similar to the disrupted firms along these and other relevant firm characteristics. We include this matched set of firms as the third difference in our analysis.

Our empirical analysis reveals that firms that experience an operational disruption enjoy a 28% smaller increase in their risk and avoid 69% of the damage to firm value if they have implemented and credibly attested to having effective control systems. We also find that disrupted firms that do not implement control systems experience an 89% larger increase in their risk and incur 4% more damage to firm value. All but the last measure are economically material and statistically significant. These results are robust to alternative event windows, study periods, methods of calculating the dependent variables, matching processes, and regression specifications.

The impact of operational disruptions on firm risk and value is salient to operations managers seeking ways to limit the consequences of operational disruptions. Many interventions to mitigate the impact of operational disruptions that have been identified in the academic literature can come with direct operational consequences in terms of higher inventory levels, operating expenses, and capital requirements (Leone 2006, Wang et al. 2010, Tomlin and Wang 2011, Dong and Tomlin 2012), implying that firms must make choices on how to fund such interventions when resources are scarce. Fostering a better understanding of whether credible control systems can reduce the impact of disruptions on the firm's risk and value is important as it identifies a broader set of mitigation strategies available to operations managers. This can help managers achieve a better fit between their risk mitigation initiatives and their objectives and budget constraints. Further analytical and empirical research in this area is warranted, however. This includes research to understand the circumstances under which such control systems are more or less valuable, whether such systems reduce the occurrence or magnitude of disruptions, whether it is useful to extend or encourage the deployment of control systems among supply chain partners, and whether there are other benefits to control systems in the face of a disruption, including worker safety, improved operational performance, and greater resiliency.

## 2. Theory and Hypotheses

To measure the impact of credible control systems, we utilize the enforcement date of regulations related to the Public Company Accounting Reform and Investor Protection Act, also known as the SOX. After a series of notorious corporate scandals involving companies such as Adelphia, Enron, and Worldcom, the U.S. Congress passed SOX in July 2002, and the SEC formalized a set of regulatory additions and changes in response to this new legislation (refer to "Spotlight on Sarbanes-Oxley Rulemaking and Reports," http://www.sec.gov/spotlight/sarbanes-oxley.htm). The regulations of primary interest to us are those intended to comply with the assessment of internal control systems—Section 404 of SOX. A large body of scholarship has examined the effects of these regulations and found that they resulted in higher earnings quality and auditing costs for complying firms (Iliev 2010) and higher equity risk for noncomplying firms (Ashbaugh-Skaife et al. 2009).[1] Less is known about whether internal control systems incrementally influence the firm's equity risk or valuation in the aftermath of an operational disruption.

Although SEC regulations are often associated with finance or accounting functions, maintaining the control processes required by Section 404 falls to both operations and finance personnel within the firm. Howell et al. (2005, p. 30) states that the process owner "could be an operational or finance person, depending on the process. . .This further embeds a culture of accountability for internal controls throughout the organization." Adherence to Section 404 requirements also has a direct impact on a firm's operations. In a survey of over 250 companies, Ernst & Young found that over 49% undertook significant remediation of core business operations prior to attaining initial compliance and 50% deployed or intended to deploy an enterprise risk management system within one year of initial compliance (Ernst & Young 2005). From a survey of over 8,200 companies, the SEC reports that the most widely reported benefit from Section 404 compliance is improvement in the quality of the respondent company's internal control structure (73% of respondents) (Office of Economic Analysis 2009). Out of consideration for the cost and effort of compliance, a well-defined set of firms was excluded from complying with Section 404 regulations (Securities and Exchange Commission 2003).[2] Those firms were never required to comply with all of the provisions of Section 404 (Securities and Exchange Commission 2003, 2006, 2008, 2010). Our identification strategy takes advantage of this difference in compliance.

Commonly applied models of determining the value of a firm involve forecasting the firm's future stream of cash flows and discounting those cash flows using an appropriate risk-adjusted rate (Brealey et al. 2011). Such models highlight that disruptions can impact firm value by affecting the cash thrown off by the firm's operations and by changing the risk associated with those cash flows. As in the Airbus example, investors may assign greater risk to the firm if they interpret disruptions as a sign that something may be wrong with the firm's operations and/or that future disruptions are likely. There is little research, however, on how the market's response to operational disruptions is influenced by the presence of credible internal control systems. Such systems may aggravate the impact of a disruption on the firm's risk and value by constraining management's flexibility to react effectively to the disruption, slowing mitigation efforts through burdensome oversight, and encumbering the firm's resources with information requirements and compliance obligations. On the other hand, these systems may reduce the impact of a disruption on the firm's risk and value by providing the tools and visibility that facilitate a more effective deployment of company resources and affording greater assurance that sufficient controls are in place to effectively manage the firm's operations. It is unclear which effect will dominate. This leads to our empirical tests of the following paired hypotheses:

**Hypothesis 1a.** *Control systems will worsen the impact of an operational disruption on the firm's equity risk.*

**Hypothesis 1b.** *Control systems will improve the impact of an operational disruption on the firm's equity risk.*

**Hypothesis 2a.** *Control systems will worsen the impact of an operational disruption on the firm's market value.*

**Hypothesis 2b.** *Control systems will improve the impact of an operational disruption on the firm's market value.*

Despite the reports of significant investments made by firms to achieve compliance, it may be that some firms had robust internal control systems already in place prior to the enforcement of these regulations. To the extent that this is pervasive in our sample, it is a bias against finding a result. In this case, the compliance date is only measuring the effect of credible information sharing about the firm's existing control systems.

## 3. Data

### 3.1. Sample

We follow the definition of a disruption in Craighead et al. (2007, p. 132) as "unanticipated events that disrupt the normal flow of goods and materials within a supply chain and, as a consequence, expose firms within the supply chain to operational and financial risks." For instance, in a manufacturing environment, disruptions include events such as an extended unscheduled plant shutdown, a material parts shortage, or a significant transportation interruption. We identify disruptions by reviewing company press releases distributed via the *PR Newswire* and *Business Wire*. Press releases are widely recognized as a common and effective means for companies to transmit information to shareholders and other constituents in a timely fashion, and their use is encouraged by U.S. securities regulators. In issuing its final rule on Regulation FD (Fair Disclosure), the SEC recommended companies use press releases as the first step in a three-step process to ensure the broad dissemination of material nonpublic information.

> We believe that issuers could use the following model, which employs a combination of methods of disclosure, for making a planned disclosure of material information, such as a scheduled earnings release. First, issue a press release, distributed through regular channels, containing the information . . . (Securities and Exchange Commission 2000, p. 51724)

The SEC observed that many companies already follow this or a similar disclosure model, and in many cases, rules of self-regulatory organizations already require companies to issue press releases to announce material developments.

To generate our sample, we search the Factiva database of press releases from January 1, 1996 to December 31, 2012. This time period was chosen in order to span the enforcement date of Section 404. This search identified announcements in which the headline or lead paragraph included terms such as *delay*, *disruption*, *interruption*, *shortage*, or *problem* in close proximity to terms like *component*, *delivery*, *parts*, *shipment*, *manufacturing*, *production*, or *operations*. A detailed documentation of the search process is available from the corresponding author. Of the approximately 7.5 million press releases in the Factiva database during our study period, the search string returned approximately 21,800 press releases. We manually reviewed and coded these announcements for relevance using two independent research assistants. The manual review process yielded 1,613 press releases of the first announcement of a disruption. Common reasons why press releases were disqualified in the manual review stage included that they pertained to a previously announced disruption or did not pertain to a disruption.

We link the information from the press releases to analysts' expectations of firm earnings from the Institutional Brokers' Estimate System (IBES) database, firm stock price information from the Center for Research in Security Prices (CRSP) database, and firm financial performance information from the Standard and Poor's COMPUSTAT database. The linking process yielded 1,345 disruption announcements from 896 firms that could be linked to these data sources. Common reasons why firms could not be linked to this requisite data included that the firm was not subject to SEC regulations, was not publicly traded, or did not file financial statements. Our primary sample consists of disruption observations and a matched set of observations for nondisrupted firms. In a robustness check, we augment our event-based analysis with time series data that includes nondisrupted periods for the disrupted firms.

Characteristics of the disrupted firms used in our analyses are reported in Table 1.

**Table 1.** Summary of Disrupted Observations Prior to Matching

| | Frequency | Percent |
|---|---|---|
| Year | | |
| 1996 | 58 | 4.3 |
| 1997 | 82 | 6.1 |
| 1998 | 92 | 6.8 |
| 1999 | 102 | 7.6 |
| 2000 | 64 | 4.8 |
| 2001 | 55 | 4.1 |
| 2002 | 48 | 3.6 |
| 2003 | 37 | 2.8 |
| 2004 | 87 | 6.5 |
| 2005 | 159 | 11.8 |
| 2006 | 85 | 6.3 |
| 2007 | 95 | 7.1 |
| 2008 | 145 | 10.8 |
| 2009 | 57 | 4.2 |
| 2010 | 82 | 6.1 |
| 2011 | 65 | 4.8 |
| 2012 | 32 | 2.4 |
| Total | 1,345 | 100 |
| Annual sales | | |
| Current year sales <$150 million | 387 | 28.8 |
| Sales ≥$150 million and <$1 billion | 349 | 26.0 |
| Sales ≥$1 billion and <$5 billion | 316 | 23.5 |
| Sales ≥$5 billion | 228 | 17.0 |
| Sales unreported | 65 | 4.8 |
| Total | 1,345 | 100 |
| Industry | | |
| Mining and Construction (SIC 1000–1999) | 258 | 19.2 |
| Food and Tobacco (SIC 2000–2199) | 30 | 2.2 |
| Textiles (SIC 2200–2399) | 14 | 1.0 |
| Lumber and Furniture (SIC 2400–2599) | 17 | 1.3 |
| Paper and Printing (SIC 2600–2799) | 29 | 2.2 |
| Chemicals and Petroleum (SIC 2800–3099) | 249 | 18.5 |
| Stone and Leather (SIC 3100–3299) | 9 | 0.7 |
| Primary and Fab. Metals (SIC 3300–3499) | 87 | 6.5 |
| Industrial Machinery (SIC 3500–3599) | 73 | 5.4 |
| Electronics (SIC 3600–3699) | 155 | 11.5 |
| Transportation Equipment (SIC 3700–3799) | 50 | 3.7 |
| Instruments (SIC 3800–3899) | 104 | 7.7 |
| Miscellaneous Mfg. (SIC 3900–3999) | 12 | 0.9 |
| Transport and Utilities (SIC 4000–4999) | 108 | 8.0 |
| Retail and Wholesale (SIC 5000–5999) | 76 | 5.7 |
| Services and Finance (SIC 6000–8999) | 68 | 5.1 |
| Other (SIC 100–999) | 6 | 0.5 |
| Total | 1,345 | 100 |

## 3.2. Measures

**3.2.1. Dependent Variables.** To test Hypotheses 1a and 1b, we use the change in volatility of a firm's daily stock returns as a measure of the change in the firm's equity risk (Armstrong and Vashishtha 2012, Gormley and Matsa 2016). We use this measure of risk (as opposed to a measure of systematic risk exposure, such as beta) because it ties more closely to the prior literature on the impact of disruptions on firm risk (Hendricks and Singhal 2005), and it aligns with the discounted cash flow model that provides a theoretical basis of our analysis. The discounted cash flow model is also captured in the Airbus example that motivates the paper. We follow Gormley and Matsa (2016) and calculate stock return volatility by taking the square root of the sum of squared daily returns over the period. To adjust for differences in the number of trading days in each period, the raw sum is divided by the number of trading days in the period. We normalize the change in each firm's stock return volatility by its prior period stock return volatility. This provides a measure of the proportional change in each firm's stock return volatility to facilitate easier comparison across firms that may have different baseline risk profiles. Using this process, we calculate volatility measures over a variety of time windows before and after the firm's disruption date—30, 90, 120, and 240 days. For instance, the measure based on 30 days uses the change in the firm's stock return volatility 30 trading days after the disruption compared with its stock return volatility 30 days before the disruption. Our primary measure of stock return volatility, *Volatility Impact*, is based on a 30-day window. We also report results based on the alternative windows.

Our dependent variable to test Hypotheses 2a and 2b, *Abnormal Return*, is created using an event study. This variable measures the difference between the actual return of the firm's stock and an estimate of the return that would have been realized had the event of interest not occurred. In our case, the event of interest is the announced disruption for disrupted firms in our sample. As described in Section 4.1, we augment our data using a matched set of nondisrupted firms. For those matched observations, we generate *Abnormal Return* corresponding to the event date of the matched disrupted firm. To conduct the event study, we utilize daily stock returns for each company, which we access through CRSP. We generate the counterfactual estimate using the market returns model summarized and described in greater detail in McWilliams and Siegel (1997).[3]

The market returns model expresses the stock return of firm $i$, for an event (i.e., a disruption announcement or the match date for nondisrupted firms) occurring on date $d$, using relative day $t$ as

$$Firm\ Return_{idt} = \eta_{id} + \theta_{id} Market\ Return_{dt} + \epsilon_{idt}. \quad (1)$$

The event date, $d$, is determined as the first trading day in which the stock market can respond to the firm's event. Thus, the event date is the date the firm announces the event if the event occurs either before the U.S. stock markets open or while the markets are open; otherwise, the event date is the following trading day. Day $t$ is an integer index that is measured relative to the event date such that $t = 0$ on the event date. For instance, $t = -1$ refers to the day prior to the event date $d$. $Firm\ Return_{idt}$ is firm $i$'s stock return on day $t$ relative to event date $d$. $Market\ Return_{dt}$ is the market return on day $t$ relative to event date $d$ using an equal-weighted portfolio of all stocks listed in the CRSP database.

To estimate Equation (1), we use ordinary least squares (OLS) with a benchmark period of 255 trading days (or approximately 1 year), ending 46 trading days prior to the event (or approximately 2 months; i.e., $t = -301, -302, \ldots, -47$). This generates estimated values $\hat{\eta}_{id}$ and $\hat{\theta}_{id}$. We apply these coefficients to actual market-return data to generate counterfactual estimates of the returns for each firm's stock under the alternative state in which the event did not occur. Abnormal returns for each day in the event window are calculated as $Abnormal\ Return_{idt} = Firm\ Return_{idt} - (\hat{\eta}_{id} + \hat{\theta}_{id} Market \times Return_{dt})$. $Abnormal\ Return_{id}$ (hereafter, *Abnormal Return*) is calculated by summing the abnormal returns over the desired number of trading days in the event window. To isolate the effect of the event, we present our results using a seven-day event window (event days −3 through 3). This event window is represented with the shorthand notation (−3, 3). We run robustness checks using alternative event windows of (−5, 5), (−4, 4), (−2, 2), and (−1, 1), and the results are not meaningfully different.

Note that our dependent variables, *Volatility Impact* and *Abnormal Return*, are measures of the percent change in the firm's risk and value (respectively) in the immediate aftermath of a disruption compared with just prior to the disruption. Using dependent variables that measure a change around an event of interest has plenty of precedents in DD and DDD empirical research (for example, Morse 2011, Tsoutsoura 2015). Using a percentage change explicitly accounts for the fact that the impact of a level change will be felt differentially by different firms based on characteristics such as firm size. For example, if a large firm with a \$10 billion market capitalization experiences a disruption and loses \$5 million in value, the relative impact is trivial. If the same loss of value occurs to a firm with a \$10 million market capitalization, the impact can be devastating. Because *Abnormal Return* from an event study is measured as a percent change in the firm's value, it addresses this misalignment. Similar logic follows for *Volatility Impact*.

**3.2.2. Independent Variables.** *Disruption* is an indicator variable that is set to "one" if the focal firm is announcing a disruption and "zero" otherwise. *Target Firm* is an indicator variable that is set to "one" if the firm is required to comply with the new controls system requirements and "zero" otherwise. *Postenforcement* is an indicator variable that is set to "one" if the event date for the firm is after enforcement of the new controls system requirements and "zero" otherwise. We assess whether a firm is in compliance with these control standards by using the auditor's opinion of the effectiveness of the firm's internal controls. This opinion is made in conjunction with auditing the firm's financial statements and is available in the COMPUSTAT database. Despite being required to comply with the control system requirements, some firms fail to comply. This is rare, occurring only five times in our data. We code these firms as *Target Firm* = 0. Our results are robust if we instead code them as *Target Firm* = 1 or drop them from the analysis.

**3.2.3. Control Variables.** We collect data on a number of control variables that prior literature has indicated may be related to the impact of disruptions on a firm's stock returns (Hendricks and Singhal 2005, Hendricks et al. 2009) and that may provide an alternative explanation for our result. For such an alternative explanation to be plausible, it must explain why there is a change in the impact of a disruption on the firm's risk and value that coincides with the enforcement period of the credible control systems requirement and why this change is different for target firms and nontarget firms.

It may be that in the postenforcement period, target firms incur or announce disruptions that are smaller in magnitude compared with those in the preenforcement period or that nontarget firms experience an increase in the economic magnitude of disruptions during the postenforcement period. To address this possibility, we include *Earnings Surprise* to control for the economic magnitude of the disruption for each firm. *Earnings Surprise* is calculated using IBES data. We subtract the average of analysts' earnings per share (EPS) forecast 10 days before the event from the average of analysts' EPS forecast 10 days after the event and divide by the firm's share price. For firms without analyst EPS forecasts, we code *Earnings Surprise* as missing using the variable *No Earnings Surprise*. There are some extreme outliers for *Earnings Surprise* (primarily because of some firms having very low share prices relative to the impact on EPS), so we winsorize this variable at 5%. This involves replacing values of *Earnings Surprise* beyond the 2.5th and 97.5th percentiles with values at the 2.5th and 97.5th percentiles. Our findings are robust if we do not winsorize *Earnings Surprise*. As a robustness check, we also generate values for *Earnings Surprise* for firms without analyst forecasts by comparing actual earnings in the disrupted quarter with a Holt–Winters forecast of earnings in that quarter. Our inferences are unchanged.

We gather additional firm-level financial information from the COMPUSTAT database and use it to calculate one-quarter lagged values for several other control variables. First, we control for the possibility that target firms systematically decrease their debt obligations in the postenforcement period relative to nontarget firms, allowing them to better weather the financial strain from disruptions that occur in the postenforcement period. To address this, we include *Debt-to-Equity Ratio* to reflect the relative amount of debt held by each firm. *Debt-to-Equity Ratio* is the book value of the firm's long-term debt divided by the market value of its common equity, and it is a measure of the amount of leverage the firm has on its balance sheet. We also include *Fixed Asset Ratio* to reflect the relative amount of liquid assets available to the firm. *Fixed Asset Ratio* is the book value of the firm's property, plant, and equipment divided by its total assets. It provides a measure of how much of a firm's capital is tied up in long-term assets and otherwise not easily accessible in a time of crisis.

Recognizing that the SOX regulations were put in place in the years following a dramatic sell-off in U.S. equity markets, it is possible that target firms suffered differentially larger reductions in their market value in the postenforcement period relative to nontarget firms. If the stock market had already incorporated bad news into the share price of target firms, it could dampen the response to disruptions in the postenforcement period. We address this by including *Market-to-Book Ratio* as a control. *Market-to-Book Ratio* is the market value of the firm's common equity divided by the book value of its common equity and indicates whether investors believe the firm is worth more or less than the book value of its equity.

Finally, it may be that there are differentially more very large target firms or very small nontarget firms in the postenforcement period compared with the preenforcement period. Such changes to the size distributions of target and nontarget firms across periods may account for differences in the market's response to disruptions across periods. To address this possibility, we also include two measures of the firm's size—log *Sales* and *Market Cap*. We also include *Market Cap* squared. log *Sales* is the natural log of the firm's quarterly sales. *Market Cap* is the market value of the firm's common equity.

Table 1 in the online appendix summarizes the variables used in our analyses.

## 4. Empirical Models

### 4.1. Matching Process

One challenge in determining how control systems affect the impact of disruptions on firm risk and value is that certain firm types may incur disruptions disproportionately in periods when there are material fluctuations in securities markets. For instance, firms in the same industry and of the same size may face disturbances in their shared consumer markets, scarcity of critical suppliers, and common dependency on raw materials. Such events may precipitate a negative stock market reaction for those firm types in general. Our interest, however, is to disambiguate the market's adverse reaction to a disrupted firm from the stock market's reaction to a troubled industry or set of firms. This possible contamination can be addressed by comparing disrupted firms with similar firms in the same time period that do not experience a disruption. We implement this utilizing coarsened exact matching.

Coarsened exact matching is a method that pairs firms experiencing a disruption with firms not experiencing a disruption based on exactly matching on coarsened values of the covariates used in the matching process. The level of coarsening for each variable is chosen so that balance is achieved across the treatment and control groups. Coarsening a variable too roughly may result in a large number of matches, but the balance test for the variable will be inadequate. Coarsening a variable too finely may result in superior balance for the variable, but it may exclude viable matches and limit the generalizability of the results. After the variables are coarsened, the disrupted and nondisrupted firms are exactly matched across all coarsened variables, and unmatched observations are pruned from the analysis. The coarsened data are then discarded, and the original covariate values are used for the balance tests and regression analyses. Iacus et al. (2012) provides a technical discussion of coarsened exact matching process.

We match based on coarsened groups of the pretreatment lagged values of both of our dependent variables, *Abnormal Returns* and *Volatility Impact*. For both variables, six coarsened groups are created corresponding to the lowest value through the 10th percentile, the 10th percentile through the 25th percentile, the 25th percentile through the 50th percentile, the 50th percentile through the 75th percentile, the 75th percentile through the 90th percentile, and the 90th percentile through the highest value. We also include the exact values of the *Target Firm* variable, the firm's two-digit standard industry classification (SIC) code, and the announcement date in our matching process. Firms not experiencing a disruption can be matched in multiple time periods. This aligns with the fact that some of the firms in our data incur multiple disruptions and may therefore be present in the data over multiple dates. Firms with a disruption during the study period are excluded from the pool of potential matched control firms. Within each coarsened group, control observations are randomly pruned to achieve one-to-one matching. As described in Section 5.4, our results are robust to alternative matching methods.

As with any matching process, a limitation of coarsened exact matching is that a contemporaneous change may occur that influences both the likelihood of being disrupted and the dependent variable, which is not captured by the variables used in the matching process. For instance, firms may experience operational disruptions contemporaneously with unrelated initiatives that impact firm-specific stock price returns and volatility. Although this is problematic in small samples, the concern is ameliorated in larger samples where such behavior is not expected to be replicated across multiple firms. As a result of the coarsened exact matching process, we generate 861 firms that announce 1,285 disruptions during our study period and 1,006 firms without disruptions covering 1,285 matched dates. *Abnormal Return* is unavailable for a small number of these observations. This accounts for the slight difference in the sample sizes used in our analysis of *Abnormal Return* compared with *Volatility Impact*. Table 2 in the online appendix provides summary statistics, and Table 3 in the online appendix provides correlations for the variables used in our analyses. Balance test results are provided in Table 2.

### 4.2. Estimating Control System Impact on Firm Risk and Value

Matching creates a valid counterfactual that can be exploited in a DDD regression analysis. In such an analysis, the nondisrupted firms serve as an unbiased benchmark of how risk and value for target and nontarget firms would have differed before and after enforcement, had those firms not been disrupted. Our DDD model subtracts this benchmark from the observed risk and value impact of target minus nontarget firms, between the pre- and postenforcement periods. Intuitively, including *Target Firm* in the DDD specification controls for time-invariant differences between the group of firms that is targeted by the control system requirement and the group of nontargeted firms. Including *Enforcement* controls for time trends common to both the target and nontarget groups. The variation that remains is the differential impact on the dependent variable of a disrupted firm compared with a nondisrupted firm, comparing target

**Table 2.** Balance Tests

| Variable | Matched observations N | Matched observations Mean | Matched observations Standard deviation | Disruption observations N | Disruption observations Mean | Disruption observations Standard deviation | Standardized difference | $p$-value |
|---|---|---|---|---|---|---|---|---|
| *Abnormal Return*, prior month | 1,282 | –0.012 | 0.0044 | 1,282 | –0.008 | 0.0054 | –0.027 | 0.496 |
| *60-Day Volatility*, lag | 1,285 | 0.036 | 0.0007 | 1,285 | 0.036 | 0.0007 | 0.032 | 0.417 |
| *90-Day Volatility*, lag | 1,285 | 0.036 | 0.0007 | 1,285 | 0.035 | 0.0006 | 0.035 | 0.370 |
| *120-Day Volatility*, lag | 1,285 | 0.036 | 0.0006 | 1,285 | 0.035 | 0.0006 | 0.034 | 0.390 |
| *240-Day Volatility*, lag | 1,285 | 0.036 | 0.0006 | 1,285 | 0.036 | 0.0006 | 0.039 | 0.321 |
| *Adjusted Inventory Turns* | 1,285 | 3.197 | 1.4565 | 1,285 | 3.938 | 2.5907 | –0.010 | 0.803 |
| *Return on Assets* | 1,285 | 0.003 | 0.0067 | 1,285 | –0.003 | 0.0021 | 0.032 | 0.421 |
| *Market-to-Book Ratio* | 1,285 | 2.691 | 0.1296 | 1,285 | 2.644 | 0.1061 | 0.011 | 0.778 |
| *Sales* | 1,285 | 1,026.0 | 160.1 | 1,285 | 1,306.8 | 136.2 | –0.053 | 0.182 |

*Note.* The standardized difference is calculated as the difference in the mean values for the disrupted firms and matched set of nondisrupted firms divided by the pooled standard deviation.

**Table 3.** Estimates of the Incremental Trend in the Prematch Value of the Firm's Abnormal Return and Volatility Change Broken Down by *Postenforcement* and *Target Firm*

| | Nontarget, preenforce | Nontarget, postenforce | Target, preenforce | Target, postenforce |
|---|---|---|---|---|
| Dependent variable: *Volatility Impact* | | | | |
| $\tau \times$ *Disruption* | –0.0057 | –0.0041 | 0.0036 | –0.0004 |
| | [0.0092] | [0.0199] | [0.0071] | [0.0057] |
| Observations | 2,464 | 1,273 | 4,039 | 7,321 |
| Dependent variable: *Abnormal Return* | | | | |
| $\tau \times$ *Disruption* | –0.2905 | –0.0246 | –0.0306 | –0.0881 |
| | [0.3227] | [0.4970] | [0.1852] | [0.1022] |
| Observations | 2,424 | 1,340 | 4,102 | 7,326 |

*Note.* OLS estimates with robust standard errors clustered by firm are in parentheses.

firms versus nontarget firms, and in the postenforcement period relative to the preenforcement period.

Thus, we can test whether control systems ameliorate or exacerbate the impact of a disruption on the firm's risk by estimating the following model:

$$\begin{aligned} &\mathit{Volatility\ Impact}_{id} \\ &\quad = \alpha + \beta_1 \cdot \mathit{Disruption}_{id} + \beta_2 \cdot \mathit{Postenforcement}_{id} \\ &\qquad + \beta_3 \cdot \mathit{Target\ Firm}_{id} \\ &\qquad + \beta_4 \cdot \mathit{Disruption}_{id} \times \mathit{Postenforcement}_{id} \\ &\qquad + \beta_5 \cdot \mathit{Disruption}_{id} \times \mathit{Target\ Firm}_{id} \\ &\qquad + \beta_6 \cdot \mathit{Postenforcement}_{id} \times \mathit{Target\ Firm}_{id} \\ &\qquad + \beta_7 \cdot \mathit{Disruption}_{id} \times \mathit{Target\ Firm}_{id} \\ &\qquad \times \mathit{Postenforcement}_{id} \\ &\qquad + \gamma' Z_{id} + \mathit{Year\text{-}Quarter} + \epsilon_{id}. \end{aligned} \tag{2}$$

The unit of analysis in this model is the firm-date. For firms that experience a disruption, date $d$ represents the date of the disruption. For matched control firms (i.e., those that do not experience a disruption), date $d$ represents the match date such that the control firms are linked to the same date as their matched disrupted firm(s). *Volatility Impact* is the change in the stock volatility for firm $i$ surrounding date $d$. $\mathit{Disruption}_{id}$ identifies whether firm $i$ announced a disruption on date $d$ or not. $\mathit{TargetFirm}_{id}$ identifies whether the firm is required to comply with the new controls system requirements. $\mathit{Postenforcement}_{id}$ identifies whether date $d$ falls after the enforcement date of the new control system requirements. Our interest is in the interaction of $\mathit{Disruption}_{id}$, $\mathit{Target\ Firm}_{id}$, and $\mathit{Postenforcement}_{id}$, which captures the differential impact on the firm's risk based on whether a disruption occurs after the enforcement date for a target firm complying with the control system requirements. The vector $\mathbf{Z}_{id}$ includes the control variables coded at the firm-date level that are described in Section 3.2.3. *Year-Quarter* is a full set of year-quarter dummy variables to account for time-varying shocks that affect all observations. The $\epsilon_{id}$ is the error term.

We evaluate whether control systems influence the impact that a disruption has on the firm's market value by estimating the following model:

$$\begin{aligned} &\mathit{Abnormal\ Return}_{id} \\ &\quad = \alpha + \beta_1 \cdot \mathit{Disruption}_{id} + \beta_2 \cdot \mathit{Postenforcement}_{id} \\ &\qquad + \beta_3 \cdot \mathit{Target\ Firm}_{id} \\ &\qquad + \beta_4 \cdot \mathit{Disruption}_{id} \times \mathit{Postenforcement}_{id} \\ &\qquad + \beta_5 \cdot \mathit{Disruption}_{id} \times \mathit{Target\ Firm}_{id} \\ &\qquad + \beta_6 \cdot \mathit{Postenforcement}_{id} \times \mathit{Target\ Firm}_{id} \\ &\qquad + \beta_7 \cdot \mathit{Disruption}_{id} \times \mathit{Target\ Firm}_{id} \\ &\qquad \times \mathit{Postenforcement}_{id} \\ &\qquad + \gamma' Z_{id} + \mathit{Year\text{-}Quarter} + \epsilon_{id}. \end{aligned} \tag{3}$$

$\mathit{Abnormal\ Return}_{id}$ is the abnormal stock return of firm $i$ surrounding date $d$. The left-hand side of Equation (3) aligns with that of Equation (2).

### 4.3. Parallel Trends

The DDD specifications in Equations (2) and (3) handle common trends and level differences in the dependent variable across the treated and control firms. However, differential trends between the treated and control firms must be verified. A critical requirement of DDD specifications is that the dependent variables be parallel for treated versus control firms in all combinations of the variables of interest prior to the match (disruption) date. To confirm this, we estimate the following models for all four combinations of *Postenforcement* and *Target Firm* in the data (i.e., the two variables take the set of values (0, 0), (0, 1), (1, 0), (1, 1)):

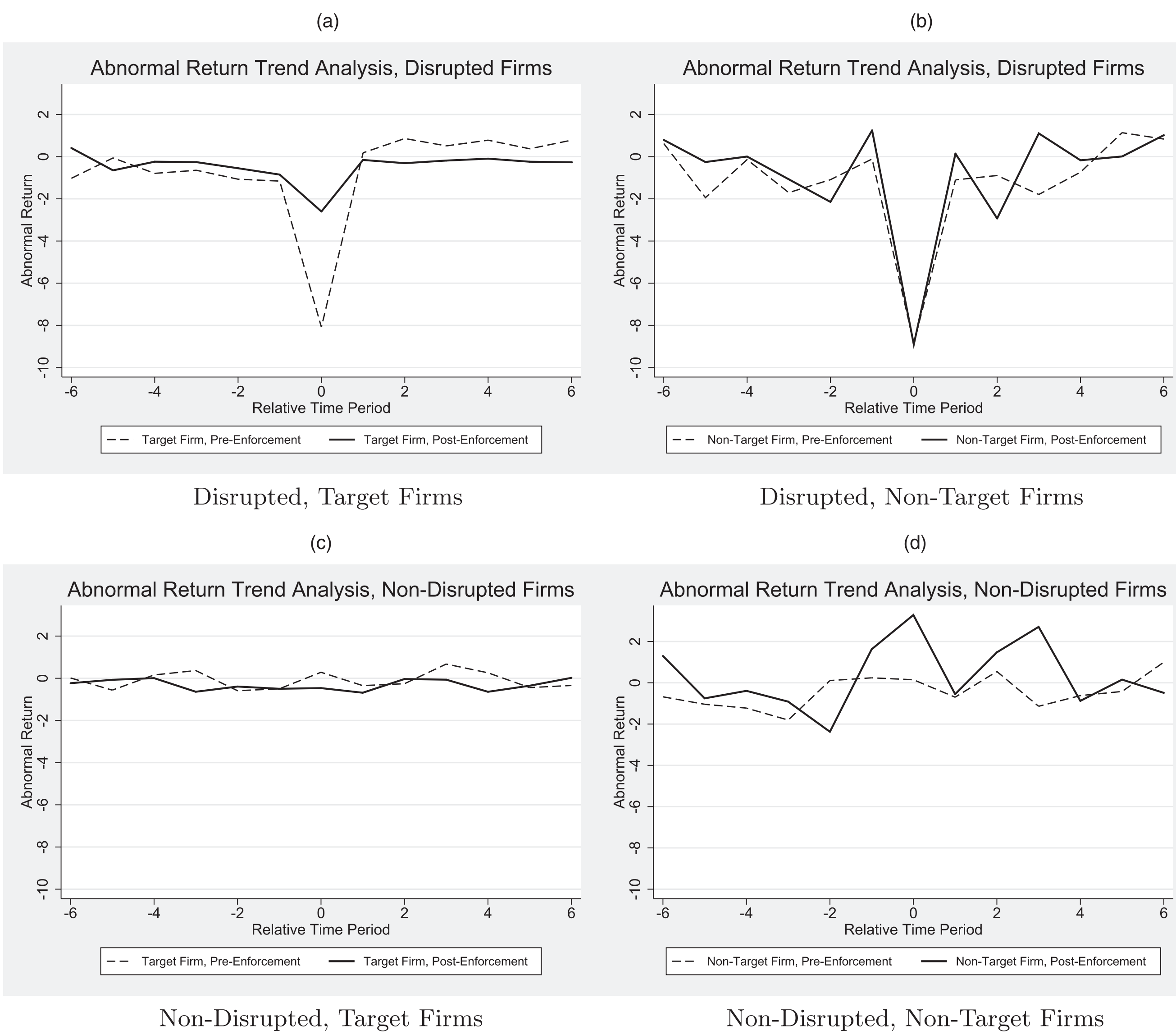
$$\begin{aligned} &\left\{\mathit{Abnormal\ Return}_{ip}, \mathit{Volatility\ Impact}_{ip}\right\} \\ &\quad = \alpha + \theta_1 \cdot \tau_{ip} \times \mathit{Disruption}_i \\ &\qquad + \theta_2 \cdot \mathit{Disruption}_i + \theta_3 \cdot \tau_{ip} + \epsilon_{ip}. \end{aligned} \tag{4}$$

We measure the dependent variables for each firm $i$ in six periods $p$ prior to each firm's match date to identify how those variables were trending prior to the match.

The length of the period varies for the two dependent variables to accommodate reasonable nonoverlapping measurement windows for the variables. The period length is 60 trading days when evaluating parallel trends for *Volatility Impact* because the dependent variable is measured using the firm's daily return volatility 30 trading days after compared with 30 days before each measurement date. The period length is seven trading days when evaluating parallel trends for *Abnormal Return* because the dependent variable uses a seven-day event window of (–3, 3). The $\tau_{ip}$ is an integer index that uniquely identifies the six periods, with $\tau_{ip} = 1$ denoting the earliest period. The coefficient $\theta_1$ represents any incremental trend present in the data for the disrupted firms relative to the matched control firms. The values of $\theta_1$ for the eight models (four combinations of *Postenforcement* and *Target Firm* for each of the two dependent variables) are presented in Table 3. None of the coefficients are material or statistically significant, providing no evidence that the parallel trends assumption is violated. Similar inferences are achieved if *Target Firm* is used in Equation (4) rather than *Disruption*.

The validity of the parallel trends assumption can also be verified visually in Figures 1 and 2. In these figures, period 1 is rescaled to be period −6 so that period 0 can represent the disruption/match date. We include period 0 and six periods following period 0 in these figures to augment our insights on the differential impact of control systems, outlined in the next section. Restricting our attention to periods −6 through −1 for the time being, Figure 1 shows that prior to the match

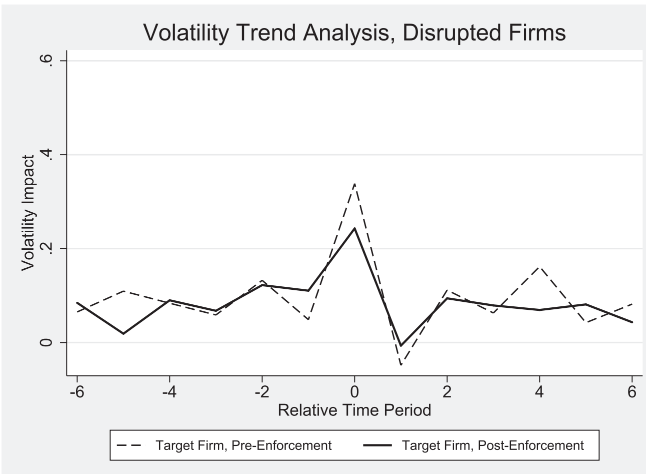
**Figure 1.** Trends in the Abnormal Stock Returns over a (–3, 3) Event Window Based on *Disruption*, *Postenforcement*, and *Target Firm*




*Notes.* (a) Disrupted target firms. (b) Disrupted nontarget firms. (c) Nondisrupted target firms. (d) Nondisrupted nontarget firms.

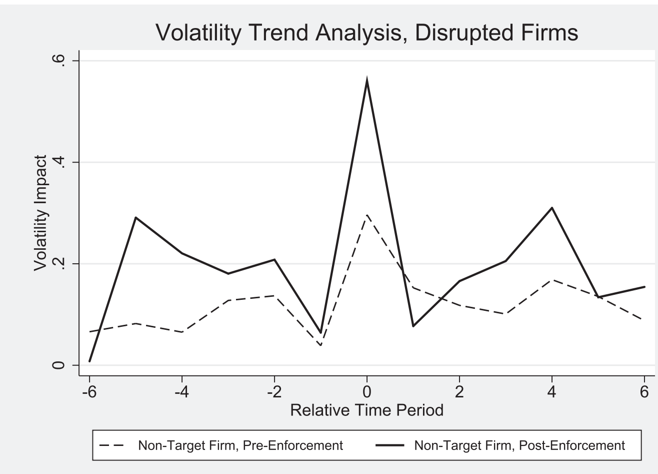
**Figure 2.** Trends in the 30-Day Volatility of the Firm's Stock Returns Based on *Disruption*, *Postenforcement*, and *Target Firm*

(a)



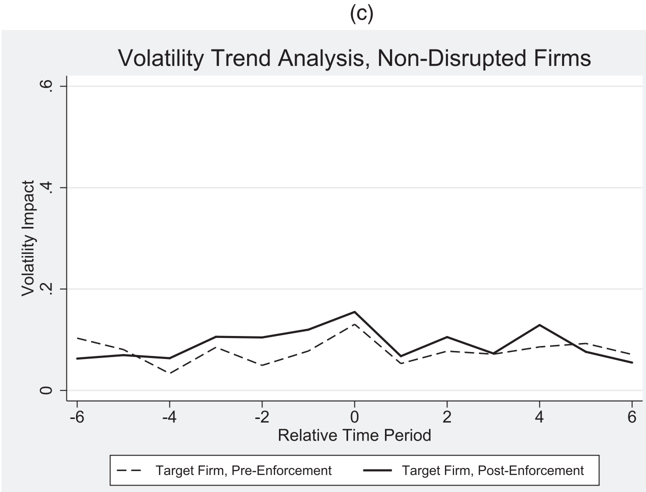

Disrupted, Target Firms

(b)



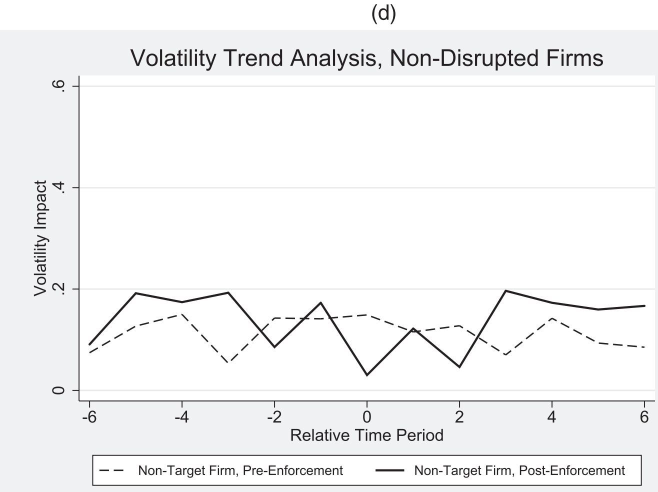

Disrupted, Non-Target Firms

(c)




Non-Disrupted, Target Firms

(d)




Non-Disrupted, Non-Target Firms

*Notes*. (a) Disrupted target firms. (b) Disrupted nontarget firms. (c) Nondisrupted target firms. (d) Nondisrupted nontarget firms.

date, *Abnormal Return* does not have a discernable differential trend for disrupted firms compared with matched firms when the data are grouped by any of the combinations of *Postenforcement* and *Target Firm*. In addition, there are not any discernable differential trends across the subfigures. For instance, the trend lines for target firms and nontarget firms are also parallel when compared across the subfigures.[4] Figure 2 shows a similar result for *Volatility Impact*.

## 5. Results

Figures 1 and 2 also provide model-free evidence that credible control systems mitigate the stock market's reaction to disruptions. In Figure 1(a), there is a substantial drop in the value of *Abnormal Return* of disrupted target firms in period 0 for the preenforcement observations (dashed line) but a much smaller drop in the value of *Abnormal Return* for disrupted target firms in the postenforcement period (solid line). In Figure 1(b), on the other hand, there is a substantial and near-identical drop in the value of *Abnormal Return* of disrupted nontarget firms in period 0 for both the preenforcement observations and the postenforcement observations. This is exactly the pattern to expect if the control systems adopted by target firms in the postenforcement period mitigated the stock market's reaction to disruptions. Figure 1, (c) and (d) does not show similar patterns in period 0 for nondisrupted firms.

In Figure 2(a), there is a substantial increase in the value of *Volatility Impact* of disrupted target firms in period 0 for the preenforcement observations but a smaller increase in the value of *Volatility Impact* for the postenforcement observations. In Figure 2(b), there is a substantial increase in the value of *Volatility Impact* of disrupted nontarget firms in period 0 for the preenforcement observations and an even larger increase in the value of *Volatility Impact* for the postenforcement observations. This pattern would be expected if the control systems adopted by target firms in the postenforcement period mitigated the stock market's reaction to disruptions and if the market treated nontarget firms as relatively *more* risky after a disruption in the postenforcement period. This latter observation was not expected, but it is informative and aligns with empirical findings that the *steady-state* risk of nontarget firms in the postenforcement period can increase (Ashbaugh-Skaife et al. 2009). We augment their findings by showing that there is a further increase in equity risk for nontarget firms in the postenforcement period following a disruption. Our DDD regression specification and proportional dependent variable measures explicitly account for differential steady-state risk across the target and nontarget firms in both the pre- and postenforcement periods. The findings in Ashbaugh-Skaife et al. (2009) underscore the need for such modeling choices. Figure 2, (c) and (d) does not show similar patterns in period 0 for nondisrupted firms.

### 5.1. Impact of Control Systems on Risk of Disrupted Firms

Next, we use our matched data to estimate the model in Equation (2) using an OLS regression with robust standard errors (SEs) clustered by firm.[5] The results, reported in Table 4, show that the increase in the firm's risk following a disruption is less pronounced for target firms in the postenforcement period compared with the preenforcement period and compared with nontarget firms in either period. Column (1) reports the regression coefficients excluding all control variables, column (2) adds control variables, and column (3) adds year-quarter fixed effects. We use the results in column (3) as our base model, but our inferences are the same among the models. It should be intuitive that in column (3), much of the impact of the *Postenforcement* variable is now absorbed by the year-quarter fixed effects.

The coefficient on *Disruption* in our base model is positive and significant ($\beta = 0.13$, SE 0.06, $p < 0.05$), meaning that following an operational disruption announcement, the firm's risk increases. This coefficient can be interpreted as contributing to an expected increase of 13% in stock return volatility following a disruption announcement. The coefficient on *Disruption* × *Postenforcement* is also positive and significant ($\beta = 0.34$, SE 0.14, $p < 0.05$), meaning that

**Table 4.** Estimation of the Impact of Announced Disruptions on the Volatility of the Firm's Stock Returns

| | Dependent variable: *Volatility Impact* | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| *Disruption* | 0.15** | 0.14** | 0.13** |
| | [0.06] | [0.06] | [0.06] |
| *Postenforcement* | −0.12** | −0.12** | 0.20 |
| | [0.06] | [0.06] | [0.14] |
| *Target Firm* | −0.02 | −0.01 | −0.01 |
| | [0.05] | [0.05] | [0.05] |
| *Disruption* × *Postenforcement* | 0.38*** | 0.39*** | 0.34** |
| | [0.14] | [0.14] | [0.14] |
| *Disruption* × *Target Firm* | 0.06 | 0.05 | 0.04 |
| | [0.08] | [0.08] | [0.07] |
| *Postenforcement* × *Target Firm* | 0.14** | 0.15** | 0.10 |
| | [0.07] | [0.07] | [0.07] |
| *Disruption* × *Postenforcement* × *Target Firm* | −0.50*** | −0.49*** | −0.43*** |
| | [0.16] | [0.15] | [0.15] |
| Control variables | No | Yes | Yes |
| Time fixed effects | No | No | Yes |
| Observations | 2,570 | 2,570 | 2,570 |
| Number of firms | 1,867 | 1,867 | 1,867 |
| Number of disruptions | 1,285 | 1,285 | 1,285 |
| $R^2$ | 0.02 | 0.03 | 0.13 |

*Notes*. The sample includes disrupted firms and a matched set of firms that are not disrupted as described in Section 4.1. OLS estimations with robust standard errors clustered by firm are in brackets. Included controls are described in Section 3.2.3.

**$p < 0.05$ for two-tailed tests; ***$p < 0.01$ for two-tailed tests.

following an operational disruption announcement in the postenforcement period, the firm's risk increases even further regardless of whether the firm is obligated to comply with the control system requirements. Finally, the coefficient on *Disruption* × *Postenforcement* × *Target Firm* is negative and significant ($\beta = -0.43$, SE 0.15, $p < 0.01$). For target firms, this last term eliminates all of the increased risk associated with disruptions in the postenforcement period and much of the increase in the risk associated with disruptions overall. As a result, target firms have a smaller proportional increase in their risk following a disruption in the postenforcement period than in the preenforcement period and a smaller proportional increase than nontarget firms in the postenforcement period. Recall that our dependent variable is a measure of the increase in the firm's risk relative to the steady-state (i.e., predisruption) risk of the firm and that by using such a measure, we account for the fact that some firms may be inherently riskier than other firms.

The actual estimated value of *Volatility Impact* for a firm between the preenforcement and postenforcement periods depends, of course, on all of the variables in the model. We conduct a marginal effects analysis to estimate the value of *Volatility Impact* by accounting for all of the data in the sample applied to *all* of the coefficients in the model (for details on marginal effects analysis, see Williams 2012). This allows for a cleaner comparison of two hypothetical disrupted, target firms that are otherwise alike in every way except that one is in the preenforcement period and the other is in the postenforcement period. The results are summarized in Table 5, Dependent variable: *Volatility Impact*. The estimated impact on *Volatility Impact* for a target firm in the preenforcement period is 0.338, whereas it is 0.243 for the firm in the postenforcement period. This means that the stock market reaction to a disruption announcement is 0.095-points lower (i.e., a lower increase in risk) if the firm has implemented and credibly communicated effective control systems. This impact of internal control systems corresponds to an economically material 28% improvement (i.e., smaller increase) in the average estimated impact of a disruption on the firm's proportional change in its stock return volatility. Our results provide support for Hypothesis 1b. This table also shows the marginal effect of the enforcement of the control system requirements for nontarget firms. There is a substantial increase in the risk of nontarget firms in the postenforcement period in the aftermath of a disruption.

### 5.2. Event Study Results

Table 6 provides the event study results for the focal event window (−3, 3), as well as the four alternative event windows (−5, 5), (−4, 4), (−2, 2), and (−1, 1). As shown in column (1), the abnormal return results are consistent for disruption observations across all time windows. The abnormal return measures for the set of disrupted firms are all negative, of a similar magnitude, economically material, and statistically significantly different from zero. As shown in column (4), the abnormal return measures for the set of matched control observations are also consistent across all time windows—economically small, of a similar magnitude, and statistically insignificant. We use two-sample $t$ tests to test for the differences in means between the disrupted observations and the matched control observations assuming unequal variances. As shown in column (7), these differences in means are all negative, of a similar magnitude, economically material, and statistically significantly different from zero.

In event studies with long event windows, parametric $t$ tests can be prone to cross-sectional correlation and event-induced volatility. Although short event windows such as the ones that we employ are much less susceptible to such issues, we nonetheless calculate the Wilcoxon rank-sum test for two independent samples (Wilcoxon 1945) as a robustness check on our parametric test results (not tabled). The rank-sum test is a nonparametric test of the null hypothesis that the abnormal returns from disrupted observations and the abnormal returns from the matched set of nondisrupted control observations are drawn from populations with the same distribution. The resulting $z$ statistic is statistically significant at a $p < 0.01$ for all event windows, providing support that there is a

**Table 5.** Marginal Effects of the Enforcement Period for Control Systems Requirements on *Volatility Impact* and *Abnormal Return* of Target and Nontarget Disrupted Firms.

| | Preenforcement | Postenforcement | Marginal effect | Change, % |
|---|---|---|---|---|
| Dependent variable: *Volatility Impact* | | | | |
| Target firm | 0.338*** | 0.243*** | −0.095** | −28.2 |
| Nontarget firm | 0.296*** | 0.560*** | 0.264** | 89.0 |
| Dependent variable: *Abnormal Return* | | | | |
| Target firm | −8.14*** | −2.55*** | 5.59*** | 68.7 |
| Nontarget firm | −8.80*** | −9.11*** | −0.31 | 3.5 |

**$p < 0.05$ using two-tailed tests; ***$p < 0.01$ using two-tailed tests.

**Table 6.** Event Study Results Broken out Between Observations with Disruptions and the Matched Set of Control Observations Without Disruptions for Event Windows (–1, 1), (–2, 2), (–3, 3), (–4, 4), and (–5, 5)

| | Disruptions | | | Controls | | | Difference | |
|---|---|---|---|---|---|---|---|---|
| | Mean | Standard error | *N* | Mean | Standard error | *N* | Mean | Standard error |
| Window (–1, 1) | –4.77*** | 0.33 | 1,278 | 0.13 | 0.33 | 1,278 | –4.90*** | 0.46 |
| Window (–2, 2) | –5.10*** | 0.37 | 1,278 | 0.43 | 0.37 | 1,278 | –5.53*** | 0.53 |
| Window (–3, 3) | –5.59*** | 0.41 | 1,278 | 0.37 | 0.40 | 1,278 | –5.97*** | 0.58 |
| Window (–4, 4) | –5.78*** | 0.44 | 1,278 | 0.49 | 0.43 | 1,278 | –6.27*** | 0.62 |
| Window (–5, 5) | –5.72*** | 0.46 | 1,278 | 0.41 | 0.43 | 1,278 | –6.13*** | 0.63 |

*Note.* Robust standard errors.
***$p < 0.01$.

difference in the distribution of abnormal returns for the underlying population of disrupted observations compared with the population of matched nondisrupted firms.

### 5.3. Impact of Control Systems on the Market Value of Disrupted Firms

We estimate the model in Equation (3) using OLS regression with robust standard errors clustered by firm. We show that the market responds less negatively to announced disruptions for target firms in the postenforcement period compared with target firms in the preenforcement period or nontarget firms in either period. The results using abnormal returns calculated over a (–3, 3) event window are presented in Table 7. Column (1) reports the regression coefficients excluding all control variables, column (2) adds control variables, and column (3) adds year-quarter fixed effects. We use the results in column (3) as our base model. Our inferences on our variables of interest are the same across all of the models.

In our base model, the coefficient on *Disruption* is negative and significant ($\beta = -8.79$, SE 1.57, $p < 0.01$). This result is in line with existing literature, which finds that operational disruptions result in a statistically significant and material reduction in firm value.

**Table 7.** Estimation of the Impact of Announced Disruptions on Abnormal Stock Returns over a (–3, 3) Event Window

| | Dependent variable: *Abnormal Returns* | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| *Disruption* | –9.39*** | –8.93*** | –8.79*** |
| | [1.57] | [1.57] | [1.57] |
| *Postenforcement* | 2.62 | 2.80 | –3.16 |
| | [3.28] | [3.31] | [5.18] |
| *Target Firm* | –0.04 | 0.25 | 0.46 |
| | [1.25] | [1.32] | [1.34] |
| *Disruption* × *Postenforcement* | –2.93 | –3.29 | –3.39 |
| | [4.11] | [4.11] | [4.03] |
| *Disruption* × *Target Firm* | 0.69 | 1.03 | 1.01 |
| | [1.89] | [1.89] | [1.91] |
| *Postenforcement* × *Target Firm* | –3.50 | –3.64 | –3.77 |
| | [3.34] | [3.35] | [3.37] |
| *Disruption* × *Postenforcement* × *Target Firm* | 9.41** | 9.12** | 9.18** |
| | [4.29] | [4.30] | [4.23] |
| Control variables | No | Yes | Yes |
| Time fixed effects | No | No | Yes |
| Observations | 2,556 | 2,556 | 2,556 |
| Number of firms | 1,857 | 1,857 | 1,857 |
| Number of disruptions | 1,278 | 1.278 | 1,278 |
| $R^2$ | 0.06 | 0.08 | 0.11 |

*Notes.* The sample includes disrupted firms and a matched set of firms that are not disrupted as described in Section 4.1. OLS estimations with robust standard errors clustered by firm are in brackets. Included controls are described in Section 3.2.3.
**$p < 0.05$ for two-tailed tests; ***$p < 0.01$ for two-tailed tests.

Unlike in the estimation of *Volatility Impact*, in the estimation of *Abnormal Return* the coefficient on *Disruption* × *Posetnforcement* is insignificant ($\beta$ = −3.39, SE 4.03, $p > 0.10$). Although this coefficient is not central to testing our hypothesis, it is arguably economically material. Finally, the coefficient on *Disruption* × *Postenforcement* × *Target Firm* is positive and significant ($\beta$ = 9.18, SE 4.23, $p < 0.05$), showing that target firms in the postenforcement period alleviate much of the damage that disruptions bring to the firm's value.

We again conduct a marginal effects analysis to estimate value of *Abnormal Return* accounting for all of the data in the sample applied to *all* of the coefficients in the model. The results are summarized in Table 5, Dependent variable: *Abnormal Return*. The resulting estimated impact on *Abnormal Return* for a disrupted target firm in the preenforcement period is −8.14, whereas it is −2.55 for a disrupted target firm in the postenforcement period. This means that the stock market reaction to a disruption announcement is 5.59 percentage points less negative (i.e., less value destruction) if the firm has implemented and attests to having effective control systems. This impact of internal control systems corresponds to an economically material 69% improvement (i.e., smaller decrease) in the average estimated impact of a disruption on the firm's abnormal stock returns. As a point of reference on the relative magnitude of this impact, the median daily return for all stocks in the CRSP database from January 1, 1991 until December 31, 2012 is 0.01 percentage points. These results provide support for Hypothesis 2b. This table also shows the marginal effect of the enforcement of the control system requirements for nontarget firms. There is little difference in the impact of disruptions on the value of nontarget firms in the postenforcement period compared with the preenforcement period.

## 5.4. Robustness Checks

We confirm that our results are robust to alternative modeling assumptions and explanations, including alternative volatility periods, event study windows, subsets of the observations in our data, regression specifications, and matching processes.

**5.4.1. Alternative Volatility Periods.** To guard against the possibility that our results are an artifact of the measurement period that we chose to estimate the change in the firm's volatility, we also consider 90 and 120 days as alternative measurement periods of *Volatility Impact*. The results are presented in columns (2) and (3) of Table 4 in the online appendix. The results are very similar to our main findings for both of these alternative windows. The coefficients on *Disruption* and *Disruption* × *Postenforcement* are largely offset by the coefficient on *Disruption* × *Postenforcement* × *Target Firm*. The interpretation of these results is consistent with the interpretation of the results from our base model. To gauge the longer-term impact of control systems on the risk induced by operational disruptions, we also use 240 trading days (or approximately one calendar year) to estimate *Volatility Impact* in column (4). Here, again we see the same pattern to the coefficients. This indicates that the beneficial impact of control systems on the firm's risk can persist long after the disruption is announced.

**5.4.2. Alternative Abnormal Return Windows.** To confirm that our results are not driven by the event window that we employ to calculate *Abnormal Return*, we run the analysis using different event windows—(−1, 1), (−2, 2), (−4, 4), and (−5, 5). These result are presented in Table 5 in the online appendix alongside the results for event window (−3, 3). In all cases, our inferences are unchanged. We also considered different estimation periods for the calculation of *Abnormal Return* (200, 150, and 100 days) and a different estimation model (the Fama–French five-factor model). In all of these cases, we generate results with similar inferences to our main results (not tabled).

**5.4.3. Public Float.** As described in Section 2, the firm's public float (i.e., the market value of its public equity) is the major determinant of whether a firm has to comply with the control system requirements (i.e., whether *Target Firm* = 1). Firms with a public float above $75 million must meet these requirements, whereas firms below this threshold are excluded from doing so. It may be that the beneficial impact on stock return volatility is primarily because of the firm's public float such that the impact of a disruption on the risk of a firm with a high public float may be inherently lower than on firms with a low public float. Although we control for this possibility with our DDD specification, as a further robustness check, we test our results by trimming firms from our analysis that are far above the $75 million threshold.

We estimate Equations (2) and (3) using firms that fall below different thresholds for public float. The results are summarized in Table 6 in the online appendix for the estimation of *Volatility Impact* and in Table 7 in the online appendix for the estimation of *Abnormal Return*. In both tables, column (1) reports the regression coefficients for our full sample, column (2) reports results for firms with a public float between $25 and $125 million, column (3) reports results for firms with a public float less than $300 million (i.e., microcap firms), column (4) reports results for firms with a public float less than $500 million, and column (5) reports results for firms with a public float less than \$1 billion (i.e., small-cap firms). Our inferences on the positive impact of controls systems on the volatility of a disrupted firm's stock returns hold for each range of public float. The results in column (2) are based on the most restrictive range for public float. Despite losing 84% of the observations in our sample, our inferences are still valid.[6]

**5.4.4. Regression Specification.** To confirm whether our results are robust to our chosen specification model, we drop our matched control observations and use a DD specification for disruptions observations only: $\textit{Volatility Impact}_{id} = \alpha + \beta_1 \cdot \textit{Postenforcement}_{id} + \beta_2 \cdot \textit{Target Firm}_{id} + \beta_3 \cdot \textit{Postenforcement}_{id} \times \textit{Target Firm}_{id} + \gamma' Z_{id} + \textit{Year-quarter} + \epsilon_{id}$. As shown in column (1) of Table 8 in the online appendix, the coefficient on *Postenforcement* × *Target Firm* is negative, statistically significant, and economically material ($\beta = -0.33$, SE 0.13, $p < 0.05$), meaning that target firms that implement credible control systems by the postenforcement period enjoy a lower *Volatility Impact* in the aftermath of a disruption compared with firms that do not implement such systems.

We use the same DD regression specification to estimation the impact on *Abnormal Return*. As shown in column (1) of Table 9 in the online appendix, the coefficient on *Postenforcement* × *Target Firm* is positive, statistically significant, and economically material ($\beta = 5.19$, SE 2.53, $p < 0.05$), meaning that target firms that implement credible control systems by the postenforcement period enjoy a materially better impact on firm value from a disruption compared with firms that do not implement such systems.

**5.4.5. Alternative Matching.** As a robustness check, we also employ alternative matching processes, including propensity score matching and Mahalanobis distance matching using uncoarsened values of the same variables used in the coarsened exact matching procedure. We include three varieties of propensity score matching that use up to 5, 10, and 20 nearest neighbors control observations. As shown in columns (2)–(5) of Table 8 in the online appendix, the coefficient on *Disruption* × *Postenforcement* × *Target Firm* for each alternative matching process is negative, economically material, and statistically significant. These results provide consistent support for Hypothesis 1b.[7] Columns (2)–(5) of Table 9 in the online appendix show consistent support for Hypothesis 2b that control systems ameliorate the impact of disruptions on *Abnormal Return*.

**5.4.6. Firm Fixed Effects.** Our dependent variables, *Abnormal Returns* and *Volatility Impact*, are forms of first differencing. Despite this, it may be that our results are driven by some firm-level fixed effects that survive. To account for this possibility, we run a robustness check that includes data on observations one month prior to the disruption and one month after the disruption. Based on this expanded data set, we include firm fixed effects in our estimation of Equations (2) and (3). In this analysis, $\textit{Disruption}_{id}$ differentiates between the disruption and nondisruption observations for each disrupted firm. We drop the matched set of nondisrupted firms because the firm's nondisrupted periods now serve as a counterfactual. The results, presented in Table 10 in the online appendix, support our findings.

**5.4.7. Market Capitalization.** We control for all of the two-way and three-way interaction effects between *Market Cap*, *Disruption*, and *Postenforcement*. The results, provided in Table 11 in the online appendix, are nearly identical to our main results.

## 5.5. Alternative Explanations

**5.5.1. Placebo Test.** To instill greater confidence that our results are not an artifact of our research design, we execute placebo tests in a context where we expect no effect. In particular, we test for an improvement in abnormal returns and the volatility of returns for target firms in the postenforcement period in randomly selected periods that did not experience a disruption. To execute this test, we drop all disrupted and matched control observations from our data. Using the remaining observations, we randomly select firms that are in the same industries, are of similar size, have the same regulatory compliance status, and had similar prior period stock return profiles. We assign a phantom disruption to these firms on the same date as the original set of disrupted observations. In some instances, such a similar observation was not available. This led to a slightly lower sample size for our placebo tests. We use the same matching and empirical estimation process on the resulting set of placebo observations. As shown in Table 12 in the online appendix, there is no evidence that controls systems provide a benefit for *Volatility Impact* or *Abnormal Return* in the aftermath of a fictional disruption. In the estimation of *Volatility Impact* in column (1), the coefficient on *Disruption* × *Postenforcement* × *Target Firm* is economically small, statistically insignificant, and *positive* (the opposite sign that we find in our main analysis). In the estimation of *Abnormal Return* in column (2), the coefficient on *Disruption* × *Postenforcement* × *Target Firm* is economically small, statistically insignificant, and *negative* (again, the opposite sign that we find in our main analysis).

**5.5.2. Economic Severity of Disruptions.** It may be that firms with credible control systems in place are simply incurring or announcing a greater number of less damaging disruptions (or suppressing announcements of more damaging disruptions) than firms without such systems. Although unlikely, such a situation would distort our sample of observable disruptions in such a way as to bias our results. We test this alternative explanation by evaluating whether *Earnings Surprise* differs between disrupted firms with credible control systems and those without credible control systems. Recall that *Earnings Surprise* identifies the earnings impact associated with the disruption and is our proxy for the economic magnitude of the disruption. We estimate $Earnings\ Surprise = \alpha + \beta_1 \cdot Disruption_{id} + \beta_2 \cdot Postenforcement_{id} + \beta_3 \cdot Target\ Firm_{id} + \beta_4 \cdot Disruption_{id} \times Postenforcement_{id} + \beta_5 \cdot Disruption_{id} \times Target\ Firm_{id} + \beta_6 \cdot Postenforcement_{id} \times Target\ Firm_{id} + \beta_7 \cdot Disruption_{id} \times Target\ Firm_{id} \times Postenforcement_{id} + \gamma' Z_{id} + Year\text{-}Quarter + \epsilon_{id}$. Vector $\boldsymbol{Z}$ is a set of control variables dimensioned by firm-date—*Debt-to-Equity Ratio*, *Market-to-Book Ratio*, *Fixed Asset Ratio*, log *Sales*, *Market Cap*, and *Market Cap* squared. Table 13 in the online appendix shows that disruptions overall are associated with a very negative earnings surprise—the coefficient on *Disruption* is negative, economically material, and statistically significant ($\beta = -0.000155$, SE 0.000032, $p < 0.01$), as is the coefficient on the coefficient on *Disruption* × *Target Firm* ($\beta = -0.000151$, SE 0.000044, $p < 0.01$). The coefficient on *Disruption* × *Target Firm* × *Postenforcement*, however, is much smaller and statistically insignificant. If disrupted firms that complied with the control system requirements in the postenforcement period had systematically announced materially less damaging disruptions after the enforcement date, this coefficient would instead be economically large and statistically significant.

## 6. Discussion and Managerial Implications

Our research investigates whether the increase in firm risk and the destruction of firm value that often accompanies operational disruptions can be ameliorated by implementing and credibly communicating the presence of effective internal control systems. We present abundant evidence that this is true. Our analysis is in the context of stock market reactions to disruptions because stock markets provide rich archival data for analyzing decisions and risk under information asymmetry. We argue, however, that our insights apply more generally to the reactions of other firm constituents in the presence of information asymmetry, including suppliers, employees, partners, customers, and regulators.

We note that our matching, DDD analysis framework, and controls may not fully resolve all conceivable sources of endogeneity. In particular, there may be some omitted variable that (1) leads to both a disruption and a differential market reaction and (2) depends on whether firms have control systems in place in the postenforcement period. Lacking a theoretical justification for such a variable, we turned to identifying a valid instrument for possible use in an instrumental variable analysis. This effort was unsuccessful because we could not empirically identify a variable that was associated with the *occurrence* of a supply chain disruption in our data. We acknowledge our unsuccessful search for an instrument in the event that future scholars are able to resolve this issue.

We also conducted an out-of-sample analysis on disruptions emanating from Canadian firms. Our window of analysis was shortened because of confounding Canadian regulations. This fact and the smaller economy of Canada relative to the United States lead to a smaller sample size. As summarized in Table 14 in the online appendix, there is no evidence that Canadian firms were impacted by the enforcement of the control system regulations that affected U.S. firms. We stress that this preliminary analysis does not rise to the level of evidentiary support for or against our hypothesized results. We include it in our paper, however, in the hope that it may motivate future research in this area. Such research may include developing a group of synthetic control firms based on a composite set of firms spanning multiple countries that did not implement SOX-like regulations. This would involve a significant amount of time and expense as press releases would have to be reviewed in the native languages of the host countries. Further complicating this is that many developed countries adopted some form of control system regulations, but enforcement and exclusion conditions were very uneven. This requires a level of expertise in the laws and regulatory processes of foreign countries that is not readily available to us. We leave this effort to future research.

A variety of mechanisms may inform the operational drivers of our results. Testing for the presence of these mechanisms requires more detailed qualitative and quantitative data than the firm-quarter level data that are available to us. Collecting data with greater detail on both the unit of observation (product level, factory level, production line level, and possibly shift level) and the time period will be difficult, however. The random and infrequent nature of large operational disruptions makes it challenging to hand collect such detailed data for a sufficient number of observations. Assuming that such data problems can be overcome, future research could address potential mechanisms, including organizational confidence in the firm's operational and disruption response processes, the level of supplier and customer engagement in those processes, the amount of production and service delivery flexibility associated with the firm's product and service lines, the impact of specific control systems, and the value of expanding control systems beyond the boundaries of the firm. We hope that our findings on the benefits of control systems in the face of disruptions will foster additional research on these topics.

We shared our research findings with managers and executives at several firms to gain their insight into some of the underlying mechanisms that may contribute to control systems being beneficial in the aftermath of a disruption. We learned that in the absence of control systems, a culture of cutting corners can take root. This can lead to a variety of unfavorable operational practices, including a "shop floor" mentality in which problems are handled in an ad hoc way, low levels of institutional learning, reduced visibility into service delivery and manufacturing processes, and a lack of systematized readiness for problem resolution. It is clear from these descriptions that such issues can be problematic in the aftermath of a disruption. The insights from our discussions are echoed in practitioner publications. For instance, American Production and Inventory Control Society (APICS), a trade association of supply chain management professionals, and Proviti, a global risk compliance consulting firm, examine the impact of SOX compliance on a firm's operations and provide several practical examples of "the effect of internal control failures on operations" (Protiviti and APICS 2003). These include inaccurate inventory balances, reduced product quality, and poor manufacturing yields. They note that in the absence of sounds control systems, there will exist a "lack of trust in the formal system" and "an environment where cutting corners became a way of life."

We identify internal control systems as an option that managers can consider to mitigate the increase in risk and loss in market value associated with disruptions. Firms may benefit by pursuing a broad range of control system investments, including supplier site audits, independent audits of specific aspects of the firm's operations and supply chain, and subtier supplier visibility. It may also be useful to consider the adoption of established rapid response processes within the supply chain, such as the observe, orient, decide, act loop methodology. For instance, Nissan has implemented a rapid response process as part of their business continuity planning to establish clear rules of action, systems, and controls when a disruption occurs. This process calls for the formation of the Disaster Core Team (DCT) within 15 minutes of the disruption and the Emergency Management Center within 120 minutes. The DCT consists of seven core functions at Nissan, three of which are manufacturing, purchasing, and supply chain. The DCT must make its initial assessment of the crisis across all seven functions within 120 minutes. The company's executives report that the benefits/success factors of this rapid response process include the prompt establishment and empowerment of cross-functional teams, smooth communication with functions and regions, and quick decision making by Nissan leadership. This facilitates swift approvals on alternative parts and suppliers, authorization for expedited deliveries, and rapid consensus on global restart sequencing and part allocation decisions.

## Endnotes

[1] Refer also to Coates and Srinivasan (2014) for a recent multidisciplinary literature review on the subject.

[2] Domestic firms had to comply only if they (1) have a public float of at least $75 million as of the last business day of the firm's most recently completed second fiscal quarter, (2) have been subject to the Exchange Act's reporting requirements for at least 12 calendar months, and (3) have previously filed at least one annual report (Securities and Exchange Commission 2002). The effective date of a firm's status begins with their annual report after December 15, 2002, and the firm's status is reevaluated on an annual basis.

[3] We chose the market returns model to present our primary results to align with other literature in this area. As a robustness check, however, we also compute *Abnormal Return* using the Fama–French five-factor model (Fama and French 2016) and achieve very similar results (not presented). The data for the five-factor model can be accessed from Kenneth French's website (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).

[4] Note that the data are somewhat smoother for target firms compared with nontarget firms, but there is not a sustained difference over time.

[5] Our analysis does not suffer from serially correlated outcomes that can plague difference-in-differences models (Bertrand et al. 2004) primarily because we do not use multiple pre- and postperiods, and most firms appear in the data once—the average number of disruptions per firm is 1.49, 75% of disrupted firms appear in the data once, and 92% appear no more than twice.

[6] The statistical significance of our results weakens at lower thresholds of public float because of the dramatic reduction in the sample size. For instance, at a public float between $25 and $125 million, there are only 17 target firms with disruptions in the postenforcement period.

[7] The coefficient on *Disruption* × *Postenforcement*, although still positive and economically material, loses its statistical significance for two of the four matching processes. Although this does not contradict our nonhypothesized finding that the risk for nontarget firms following a disruption increases in the postenforcement period compared with the preenforcement period, it does highlight that the strength of this unexpected result is sensitive to the matching process.

## References

U.S. House of Representatives. 107th Congress (2002). 3763:Sarbanes-Oxley Act of 2002. 15 USC 7201, 107–204, Washington DC: U.S. Government Printing Office.

Airbus (2006) Airbus maintains target date for A380 certification and first delivery by year end, but announces shift in the production programme. Press Release. Accessed July 30, 2019, http://www.airbus.com/newsroom/press-releases/en/2006/06/airbus-maintains-target-date-for-a380-certification-and-first-delivery-by-year-end-but-announces-shift-in-the-production-programme.html.

Armstrong CS, Vashishtha R (2012) Executive stock options, differential risk-taking incentives, and firm value. *J. Financial Econom.* 104(1):70–88.

Ashbaugh-Skaife H, Collins DW, Kinney WR Jr, Lafond R (2009) The effect of SOX internal control deficiencies on firm risk and cost of equity. *J. Accounting Res.* 47(1):1–43.

Bertrand M, Duflo E, Mullainathan S (2004) How much should we trust differences-in-differences estimates? *Quart. J. Econom.* 119(1): 249–275.

Brealey RA, Myers SC, Allen F (2011) *Principles of Corporate Finance*, 10th ed. (McGraw-Hill/Irwin, New York).

Coates JC, Srinivasan S (2014) SOX after ten years: A multidisciplinary review. *Accounting Horizons* 28(3):627–671.

Craighead CW, Blackhurst J, Rungtusanatham MJ, Handfield RB (2007) The severity of supply chain disruptions: Design characteristics and mitigation capabilities. *Decision Sci.* 38(1):131–156.

Dada M, Petruzzi NC, Schwarz LB (2007) A newsvendor's procurement problem when suppliers are unreliable. *Manufacturing Service Oper. Management* 9(1):9–32.

Dobie G, Hubmann C, Polke-Markmann H, Vanheyden P (2018) Allianz risk barometer—top business risks 2018. Accessed April 4, 2018, http://www.agcs.allianz.com/insights/white-papers-and-case-studies/allianz-risk-barometer-2018/.

Dong L, Tomlin B (2012) Managing disruption risk: The interplay between operations and insurance. *Management Sci.* 58(10): 1898–1915.

Ernst & Young (2005) Emerging trends in internal controls fourth survey and industry insights. Accessed November 30, 2018, www.eycom.ch/publications/items/sox404/internalcontrol05/en.pdf.

Fama EF, French KR (2016) Dissecting anomalies with a five-factor model. *Rev. Financial Stud.* 29(1):69–103.

Fidler B, Memmi G (2006) A380 woes; downgrading recommendation to hold. June 13. Deutche Bank AG Report; London, England.

Gormley TA, Matsa DA (2016) Playing it safe? Managerial preferences, risk, and agency conflicts. *J. Financial Econom.* 122(3): 431–455.

Guajardo JA, Morris A Cohen S-HK, Netessine S (2012) Impact of performance-based contracting on product reliability: An empirical analysis. *Management Sci.* 58(5):961–979.

Hendricks KB, Singhal VR (2003) The effect of supply chain glitches on shareholder wealth. *J. Oper. Management* 21(5):501–522.

Hendricks KB, Singhal VR (2005) An empirical analysis of the effect of supply chain disruptions on long-run stock price performance and equity risk of the firm. *Production Oper. Management* 14(1):35–52.

Hendricks KB, Singhal VR, Zhang R (2009) The effect of operational slack, diversification, and vertical relatedness on the stock market reaction to supply chain disruptions. *J. Oper. Management* 27(3):233–246.

Hopp WJ, Spearman ML (2001) *Factory Physics*, 2nd ed. (Irwin/McGraw-Hill, New York).

Howell RA, Cheryl de Mesa G, Sinnett WM (2005) Sarbanes-OxleySection 404implementation: Practices of leading companies. Accessed April 3, 2019, http://www.grantthornton.com.

Iacus SM, King G, Porro G, Katz JN (2012) Causal inference without balance checking: Coarsened exact matching. *Political Anal.* 20(1):1–24.

Iliev P (2010) The effect of SOXSection 404: Costs, earnings quality, and stock prices. *J. Finance* 65(3):1163–1196.

Leone M (2006) A case for fatter supply chains. Accessed April 1, 2018, http://ww2.cfo.com/accounting-tax/2006/02/a-case-for-fatter-supply-chains/.

McWilliams A, Siegel D (1997) Event studies in management research: Theoretical and empirical issues. *Acad. Management J.* 40(3):626–657.

Morse A (2011) Payday lenders: Heroes or villains? *J. Financial Econom.* 102(1):28–44.

Office of Economic Analysis (2009) Study of the Sarbanes-Oxley Act of 2002Section 404internal control over financial reporting requirements. Accessed March 11, 2019, http://www.sec.gov/news/studies/2009/sox-404_study.pdf.

Protiviti; APICS (2003) Capitalizing on Sarbanes-Oxley compliance to build supply chain advantage. Accessed May 11, 2019, https://www.protiviti.com/sites/default/files/united_states/insights/supplychain_soa_0.pdf.

Securities and Exchange Commission (2000) Selective disclosure and insider trading (final rule). *Federal Register* 65(165): 51716–51740.

Securities and Exchange Commission (2002) Acceleration of periodic report filing dates and disclosure concerning website access to reports (final rule). *Federal Register* 67(179):58480–58507.

Securities and Exchange Commission (2003) Management's reports on internal control over financial reporting and certification of disclosure in exchange act periodic reports (final rule). *Federal Register* 68(117):36636–36673.

Securities and Exchange Commission (2006) Internal control over financial reporting in exchange act periodic reports of non-accelerated filers and newly public companies (final rule). *Federal Register* 71(245):76580–76599.

Securities and Exchange Commission (2008) Internal control over financial reporting in exchange act periodic reports of non-accelerated filers (final rule). *Federal Register* 73(128): 38094–38100.

Securities and Exchange Commission (2010) Internal control over financial reporting in exchange act periodic reports of non-accelerated filers (final rule). *Federal Register* 75(182): 57385–57388.

Sheffi Y (2005) *The Resilient Enterprise: Overcoming Vulnerability for Competitive Advantage* (MIT Press, Cambridge, MA).

Simchi-Levi D, Schmidt W, Wei Y, Zhang PY, Combs K, Ge Y, Gusikhin O, Sanders M, Zhang D (2015) Identifying risks and mitigating disruptions in the automotive supply chain. *Interfaces* 45(5):375–390.

Swinney R, Netessine S (2009) Long-term contracts under the threat of supplier default. *Manufacturing Service Oper. Management* 11(1): 109–127.

Tang CS (2006) Robust strategies for mitigating supply chain disruptions. *Internat. J. Logist. Res. Appl.* 9(1):33–45.

Tomlin B (2006) On the value of mitigation and contingency strategies for managing supply chain disruption risks. *Management Sci.* 52(5):639–657.

Tomlin B, Wang Y (2011) Operational strategies for managing supply chain disruption risk. Kouvelis P, Dong L, Boyabatli O, Li R, eds. *The Handbook of Integrated Risk Management in Global Supply Chains* (John Wiley & Sons, Hoboken, NJ), 79–101.

Tsoutsoura M (2015) The effect of succession taxes on family firm investment: Evidence from a natural experiment. *J. Finance* 70(2):649–688.

Wang Y, Gilland W, Tomlin B (2010) Mitigating supply risk: Dual sourcing or process improvement? *Manufacturing Service Oper. Management* 12(3):489–510.

Wilcoxon F (1945) Individual comparisons by ranking methods. *Biometrics Bull.* 1(6):80–83.

Williams R (2012) Using the margins command to estimate and interpret adjusted predictions and marginal effects. *Stata J.* 12(2): 308–331.

World Economic Forum; Accenture (2013) Building resilience in supply chains. Accessed April 11, 2018, http://www.weforum.org/reports/building-resilience-supply-chains.